# Defining Learning, Design, and Technology

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Learning, Design, and Technology is the planned work of building the conditions under which people learn. Someone in this field studies who the learners are, decides what those learners need to be able to do, and then arranges content, activities, practice, feedback, and tools so that they can get there. Technology belongs inside that arrangement, but it is not the point of it. The short version I would give at a family dinner is this: I figure out how people learn something, and then I build the thing that helps them learn it.

I define the field this way because the design decision has always mattered more than the delivery device. Reiser (2001) traces the field from early audiovisual instruction and wartime training toward a systematic process for solving instructional problems, and that move from media to process is the same move I am making in my own definition. Dale (1954) organized learning experiences by how direct or abstract they were, from doing a thing, to watching a thing, to reading about a thing. His concern was the quality of the learner's experience, not the film projector that happened to carry it. That concern is still the right one. Every few years a new tool arrives and gets described as the thing that will finally fix education, and every few years the tool turns out to be only as good as the instruction wrapped around it.

Moore (1993) gave me the second piece of my definition. He described transactional distance as the gap between what a teacher intends and what a learner understands, and he argued that the size of that gap depends on the structure of the course and the dialogue inside it rather than on physical separation. A student sitting in the third row can be far away. A student two states over can be close. If distance is really a design problem instead of a geography problem, then the job of this field is to manage that gap on purpose, in any setting.

The third piece is that tools are not neutral. Larreamendy-Joerns and Leinhardt (2006) warned that “artifacts and tools are not innocuous” (p. 597), because the practices and conversations built around a tool are what give it meaning. Shearer (2013) offers the practical version of the same idea, noting that choosing a technology means understanding how it lets the other parts of a course behave inside a system. Put together, these readings tell me that selecting a platform is already an instructional decision, and it should be made by someone who is thinking about learning.

I came into this field sideways. I write software for higher education, mostly student information systems, degree audits, and the reports that advisors and students use to decide what to take next. Before that I certified education benefits for student veterans. Both jobs taught me the same lesson: students learn from far more than their courses. A degree audit is instruction. So is a benefits letter and a registration screen. When those artifacts are confusing, students make worse decisions about their own education, and no amount of good classroom teaching repairs that. I have also worked on a peer reviewed study of an online mathematics learning platform, which showed me how much the design choices inside a system shape what students actually do inside it.

My aspiration is to keep moving toward learning engineering, where instructional design meets software development and data. I want to help build adaptive and intelligent learning systems that adjust to the learner, and I want the people building them to be able to explain their choices in terms of learning theory rather than in terms of features. I expect artificial intelligence to change the tools of this field faster than most of us are ready for. The questions underneath the tools, which are who is learning, what do they need, and how will we know it worked, will not change at all.

## References

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